

SRE CASE STUDY

AZURE ARC HYBRID CLOUD LAB

Proof-of-Value project: hybrid cloud orchestration, observability, FinOps, IaC, Policy as Code, and CI/CD pipeline.

Author: Bartosz Suszko

Company: IT Solutions Bartosz Suszko, Olsztyn, Poland

Date: May 2026 (Week 1-2)

GitHub: github.com/buth11/sre-homelab-azure-arc

Stack: K3s | Azure Arc | Prometheus | Grafana | Terraform | KQL | Azure Policy | Azure DevOps

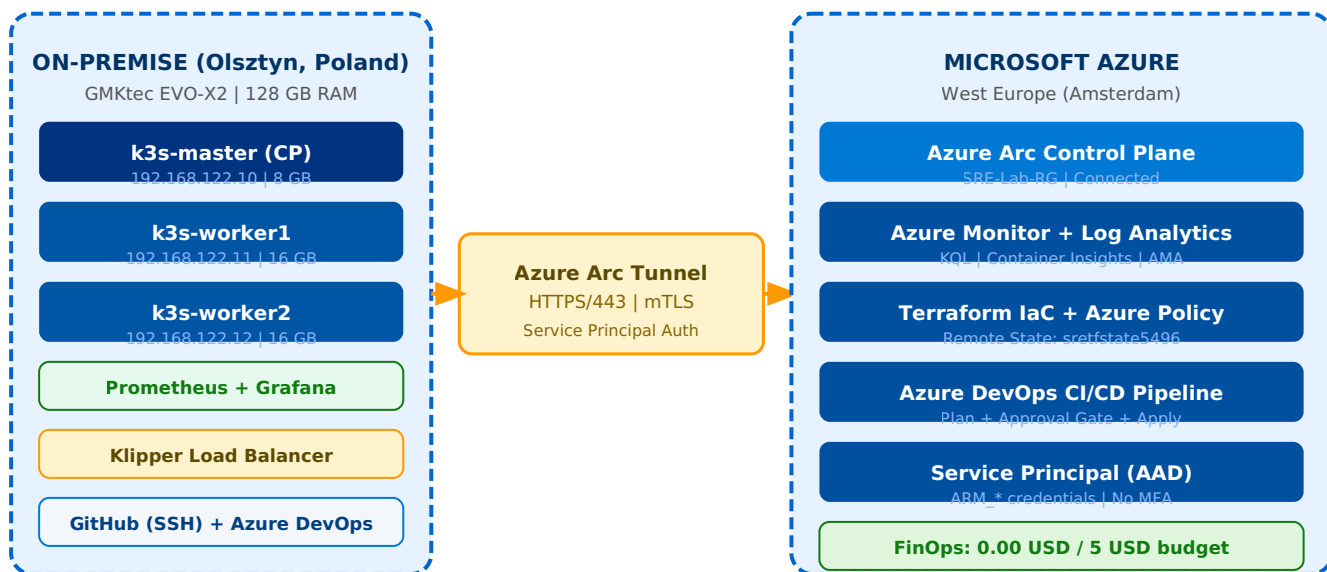
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Chapter 1: Context and Architecture

This project demonstrates building a production-grade hybrid environment on GMKtec EVO-X2 (128 GB RAM) connecting K3s Kubernetes with Microsoft Azure through Azure Arc.

Architecture Diagram (Week 1+2)



Chapter 2: On-Premise Infrastructure

Node	Role	IP	CPU	RAM
k3s-master	Control Plane	192.168.122.10	4 vCPU	8 GB
k3s-worker1	Worker	192.168.122.11	4 vCPU	16 GB
k3s-worker2	Worker	192.168.122.12	4 vCPU	16 GB

Chapter 3: FinOps

The screenshot displays the Microsoft Azure Cost Management console. The left sidebar contains navigation links for various Azure services and cost management tools. The main content area shows the 'Cost Management: Bartosz Suszko | Budżety' page. A table lists the budget details for 'SRE-Lab-Budget'.

Nazwa	Zakres	Resetuj okres	Data utworzenia	Data wygaśnięcia	Budżet	Prognostyczne	Oszacowane wy...	Postęp
SRE-Lab-Budget	55df28b8-39ed-58b...	Monthly	1.05.2026	30.04.2028	5,00 USD	0	\$0.00	0.00%

Screenshot 1: Azure Cost Management — SRE-Lab-Budget 5 USD/month, final cost 0.00 USD.

Chapter 4: Azure Arc

The screenshot shows the Microsoft Azure portal interface. The main content area displays the 'Azure Arc | Klastry Kubernetes' page for the 'K3s-Homelab' cluster. The cluster is in a 'Connected' state. The page includes a search bar, a list of resources, and a detailed view of the cluster's properties and monitoring options.

Cluster Details:

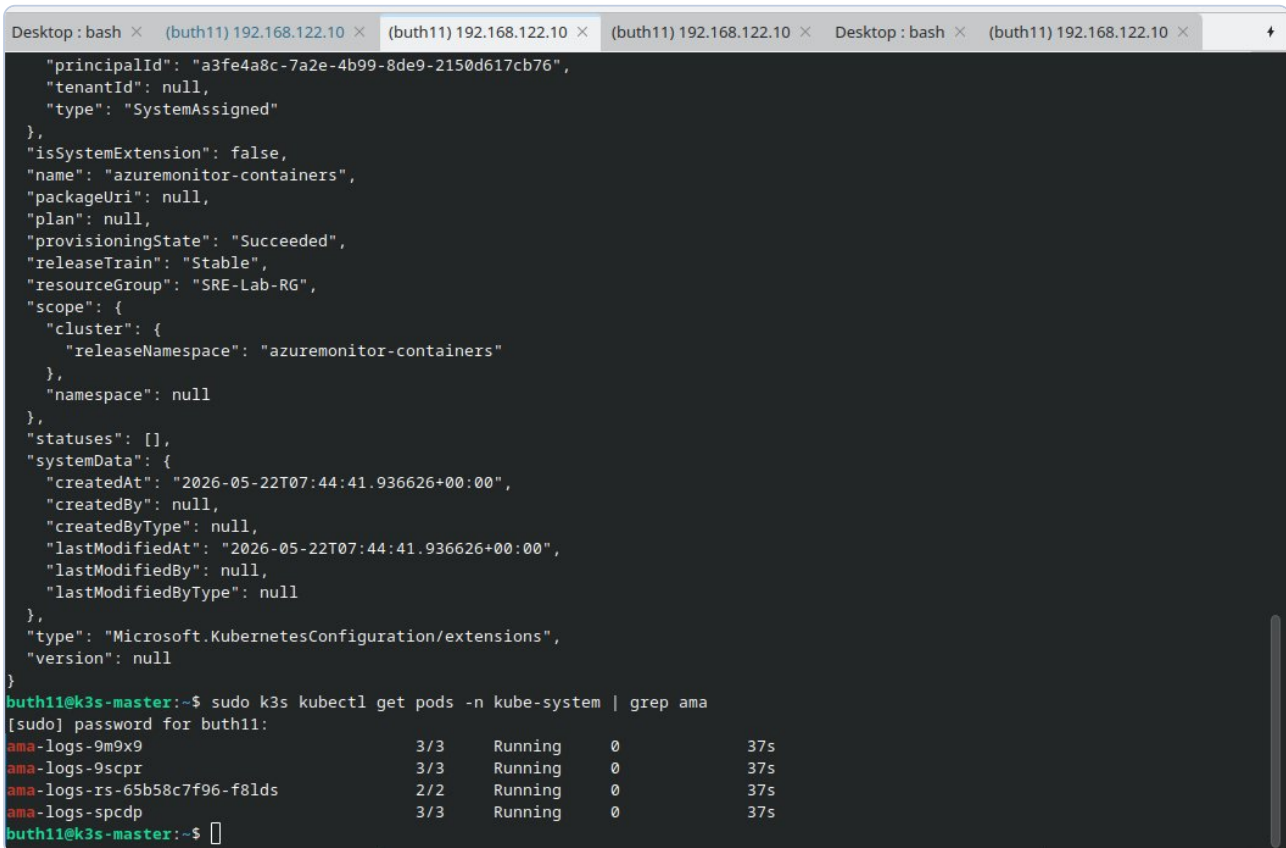
- Subskrypcja:** Azure subscription 1
- Identyfikator subskrypcji:** b023e8b8-4448-4d4f-9fcf-d20a6c1fa093
- Grupa zasobów:** SRE-Lab-RG
- Stan:** Connected
- Lokalizacja:** West Europe
- Menedżer floty:** Menedżer floty
- Wersja agenta:** 1.33.0
- Wersja platformy Kubernetes:** 1.35.4+k3s1
- Łączna liczba węzłów:** 3
- Łączna liczba rdzeni:** 12

Monitoring and Alerts:

- Właściwości:** Właściwości, Monitorowanie, Możliwości, Samouczki
- Wersja agenta:** 1.33.0
- Czas wygaśnięcia certyfikatu tożsamości zarządzanej:** August 13, 2026 at 08:20 AM GMT+2
- Wersja platformy Kubernetes:** 1.35.4+k3s1
- Łączna liczba węzłów:** 3
- Łączna liczba rdzeni:** 12

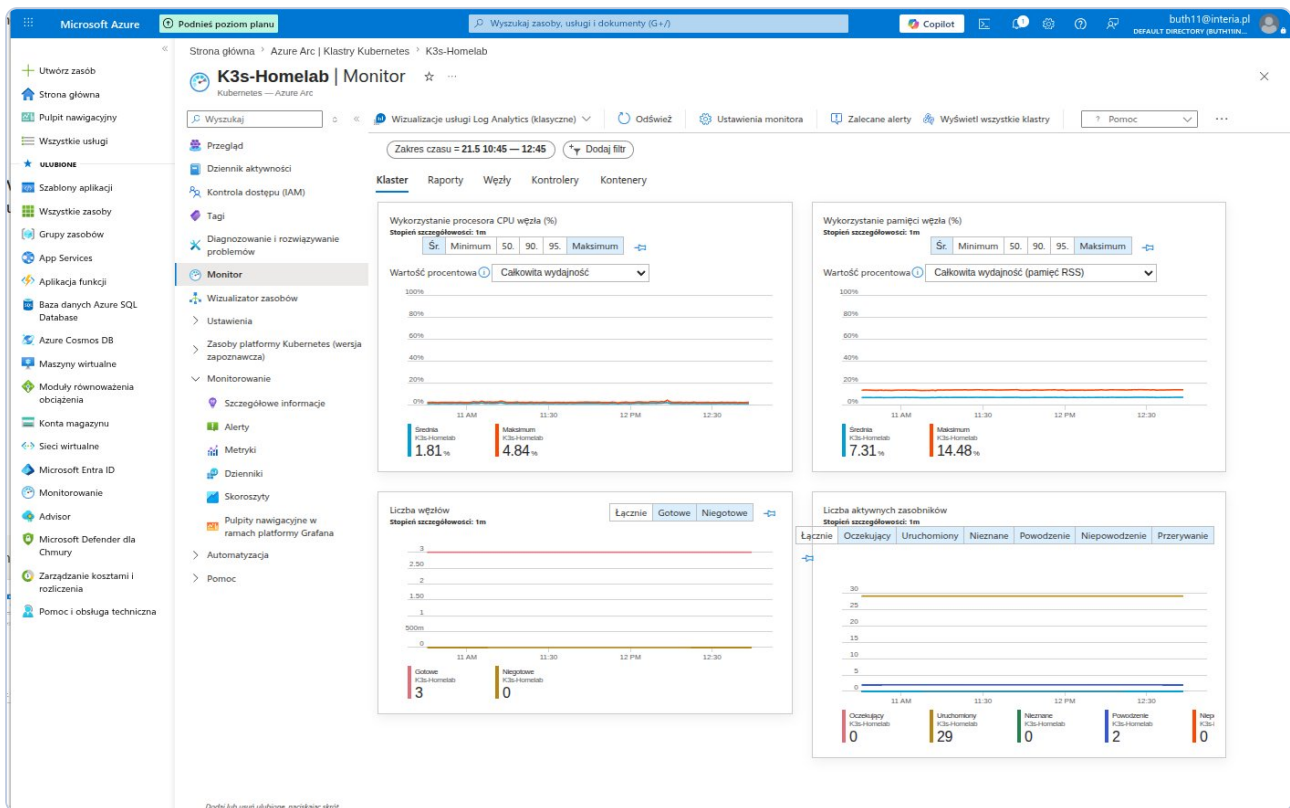
Screenshot 2: Azure Arc — K3s-Homelab Connected, v1.35.4+k3s1, 3 nodes, 12 cores.

Chapter 5: Observability



```
Desktop: bash × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × Desktop: bash × (buth11) 192.168.122.10 ×
{
  "principalId": "a3fe4a8c-7a2e-4b99-8de9-2150d617cb76",
  "tenantId": null,
  "type": "SystemAssigned"
},
"isSystemExtension": false,
"name": "azuremonitor-containers",
"packageUri": null,
"plan": null,
"provisioningState": "Succeeded",
"releaseTrain": "Stable",
"resourceGroup": "SRE-Lab-RG",
"scope": {
  "cluster": {
    "releaseNamespace": "azuremonitor-containers"
  },
  "namespace": null
},
"statuses": [],
"systemData": {
  "createdAt": "2026-05-22T07:44:41.936626+00:00",
  "createdBy": null,
  "createdByType": null,
  "lastModifiedAt": "2026-05-22T07:44:41.936626+00:00",
  "lastModifiedBy": null,
  "lastModifiedByType": null
},
"type": "Microsoft.KubernetesConfiguration/extensions",
"version": null
}
buth11@k3s-master:~$ sudo k3s kubectl get pods -n kube-system | grep ama
[sudo] password for buth11:
ama-logs-9m9x9          3/3      Running    0          37s
ama-logs-9scpr          3/3      Running    0          37s
ama-logs-rs-65b58c7f96-f8lds  2/2      Running    0          37s
ama-logs-spcdp          3/3      Running    0          37s
buth11@k3s-master:~$
```

Screenshot 3: AMA pods Running — 4 agents, 0 restarts, 37s after install.



Screenshot 4: Azure Monitor — CPU avg 1.58%, RAM avg 6.27%, 3 nodes Ready, 29 pods.

Chapter 6: KQL

The screenshot displays the Microsoft Azure portal interface. On the left is the navigation pane with options like 'Create a resource', 'Dashboard', and 'All services'. The main area shows the 'Log Analytics workspace' for 'DefaultWorkspace-b023e8b8-4446-4d4f-9fcf-d20a6c1fa093-WEU'. A 'New Query' window is open, showing a KQL query:

```
1 KubeNodeInventory
2 where TimeGenerated > ago(1h)
3 summarize arg_max(TimeGenerated, *) by Computer
4 project
5   Computer,
6   Status,
7   KubeletVersion
8 order by Computer asc
```

The query results are displayed in a table with columns: Computer, Status, and KubeletVersion. The results show 3 nodes in a 'Ready' state with 'Kubernetes v1.35.4+k3s1'.

Computer	Status	KubeletVersion
k3s-master	Ready	v1.35.4+k3s1
k3s-worker1	Ready	v1.35.4+k3s1
k3s-worker2	Ready	v1.35.4+k3s1

KQL-1: 3 nodes Ready, v1.35.4+k3s1.

Log Analytics workspace | DefaultWorkspace-b023e8b8-4446-4d4f-9fcf-d20a6c1fa093-WEU | Logs

Search resources, services, and docs (G+/)

Observability agent (New) | Save | Share | Queries hub

New Query 1* | Time range: Set in query | Show: 1000 results | KQL mode

```

1 KubePodInventory
2 where TimeGenerated > ago(1h)
3 summarize arg_max(TimeGenerated, *) by Name, Namespace
4 summarize PodCount = count() by Namespace, PodStatus
5 order by Namespace asc, PodCount desc

```

Namespace	PodStatus	PodCount
azure-arc	Running	12
default	Running	3
kube-system	Running	14
kube-system	Succeeded	2

Showing 1 - 1 of 1. Display count: au...

0s 533ms | Display time (UTC+00:00) | Query details | 1 - 4 of 4

KQL-2: azure-arc 12 Running, default 3 Running, kube-system 14+2.

Log Analytics workspace | DefaultWorkspace-b023e8b8-4446-4d4f-9fcf-d20a6c1fa093-WEU | Logs

Search resources, services, and docs (G+/)

Observability agent (New) | Save | Share | Queries hub

New Query 1* | Time range: Set in query | Show: 1000 results | KQL mode

```

1 KubeEvents
2 where TimeGenerated > ago(24h)
3 project TimeGenerated, Namespace, Name, Reason, Message
4 order by TimeGenerated desc
5 take 20

```

TimeGenerated [UTC]	Namespace	Name	Reason
5/22/2026, 7:44:56.000 AM	kube-system	ama-logs-9scpr	FailedMount
5/22/2026, 7:44:56.000 AM	kube-system	ama-logs-rs-65b58c7996-f8ld5	FailedMount
5/22/2026, 7:35:32.000 AM	azure-arc	kube-aad-proxy-5f59465457-l2...	Unhealthy
5/22/2026, 7:35:25.000 AM	azure-arc	config-agent-75548f59c5-k4zrv	Unhealthy
5/22/2026, 7:35:13.000 AM	k3s-worker2		Rebooted
5/22/2026, 7:35:11.000 AM	k3s-worker1		Rebooted
5/22/2026, 7:35:09.000 AM	kube-system	metrics-server-786d997795-zz...	Unhealthy
5/22/2026, 7:35:07.000 AM	k3s-master		Rebooted
5/21/2026, 10:53:39.000 AM	kube-system	ama-logs-89qgj	Unhealthy
5/21/2026, 10:52:38.000 AM	kube-system	ama-logs-rs-65b58c7996-nlxhw	Unhealthy
5/21/2026, 8:37:39.000 AM	kube-system	svclb-aras-demo-app-66f7352...	FailedScheduling
5/21/2026, 8:37:39.000 AM	kube-system	svclb-aras-demo-app-66f7352...	FailedScheduling
5/21/2026, 8:37:39.000 AM	kube-system	svclb-aras-demo-app-66f7352...	FailedScheduling

TimeGenerated [UTC] 2026-05-21T08:37:39Z

Namespace kube-system

Name svclb-aras-demo-app-66f7352-p9nqc

Reason FailedScheduling

Message 0/3 nodes are available: 1 node(s) didn't have free ports for the requested pod ports, 2 node(s) didn't satisfy ... s are available: 1 node(s) didn't have free ports for the requested pod ports, 2 Preemption is not helpful for sc...

Showing 1 - 1 of 1. Display count: au...

0s 287ms | Display time (UTC+00:00) | Query details | 1 - 13 of 13

KQL-3: 13 events 24h — Rebooted, Unhealthy (temporary), FailedScheduling.

Microsoft Azure | Upgrade | Search resources, services, and docs (G+/)

Home > Log Analytics workspaces > DefaultWorkspace-b023e8b8-4446-4d4f-9fcf-d20a6c1fa093-WEU

Log Analytics workspace

Default Directory (buth11@interia.onmicrosoft.com)

Log Analytics workspace

Search

New Query 1

Time range: Set in query | Show: 1000 results | KQL mode

```

1 KubePodInventory
2 | where TimeGenerated > ago(24h)
3 | summarize arg_max(TimeGenerated, *) by Name, Namespace
4 | where PodRestartCount > 0
5 | project Namespace, Name, PodRestartCount, PodStatus
6 | order by PodRestartCount desc

```

Namespace	Name	PodRestartCount	PodStatus
azure-arc	extension-manager-85899588c...	3	Running
azure-arc	clusterconnect-agent-5bc68597...	3	Running
kube-system	svcib-traffic-4b7f5cb6-88kx	2	Running
azure-arc	extension-events-collector-6c9...	2	Running
azure-arc	clusteridentityoperator-767768...	2	Running
azure-arc	kube-aad-proxy-5f53465457-42...	2	Running
kube-system	svcib-traffic-4b7f5cb6-c2gs	2	Running
azure-arc	config-agent-75548f5c5-k4zrv	2	Running
azure-arc	cluster-metadata-operator-8db...	2	Running
kube-system	svcib-traffic-4b7f5cb6-9587	2	Running
azure-arc	metrics-agent-6d994764-rcr44	2	Running
azure-arc	controller-manager-7cb948bdf...	2	Running
azure-arc	resource-sync-agent-54d88c5...	2	Running
azure-arc	flux-logs-agent-768c454c3-9g...	1	Running
azure-arc	logcollector-8bd4468b4-4s2zw	1	Running
kube-system	ama-logs-e6g5p	1	Running
kube-system	helm-install-traffic-ggts	1	Succeeded
kube-system	svcib-aras-demo-app-8581479...	1	Running
default	aras-demo-app-f78659546-vb...	1	Running
kube-system	coredns-c4dffb5f-84kx	1	Running
kube-system	local-path-provisioner-5c4dc5...	1	Running
kube-system	metrics-server-786d97795-zz...	1	Running
kube-system	svcib-aras-demo-app-8581479...	1	Running

0s 381ms | Display time (UTC+00:00) | Query details | 1 - 24 of 27

KQL-4: 27 pods with restarts after VM reboot — all Running.

Microsoft Azure | Upgrade | Search resources, services, and docs (G+/)

Home > Log Analytics workspaces > DefaultWorkspace-b023e8b8-4446-4d4f-9fcf-d20a6c1fa093-WEU

Log Analytics workspace

Default Directory (buth11@interia.onmicrosoft.com)

Log Analytics workspace

Search

New Query 1

Time range: Last 24 hours | Show: 1000 results | KQL mode

```

1 KubeNodeInventory
2 | where TimeGenerated > ago(24h)
3 | summarize
4 |   TotalSamples = count(),
5 |   ReadySamples = countif(Status == "Ready")
6 | by Computer
7 | extend AvailabilityPct = round(toreal(ReadySamples) / toreal(TotalSamples) * 100, 2)
8 | project Computer, AvailabilityPct, ReadySamples, TotalSamples
9 | order by Computer asc

```

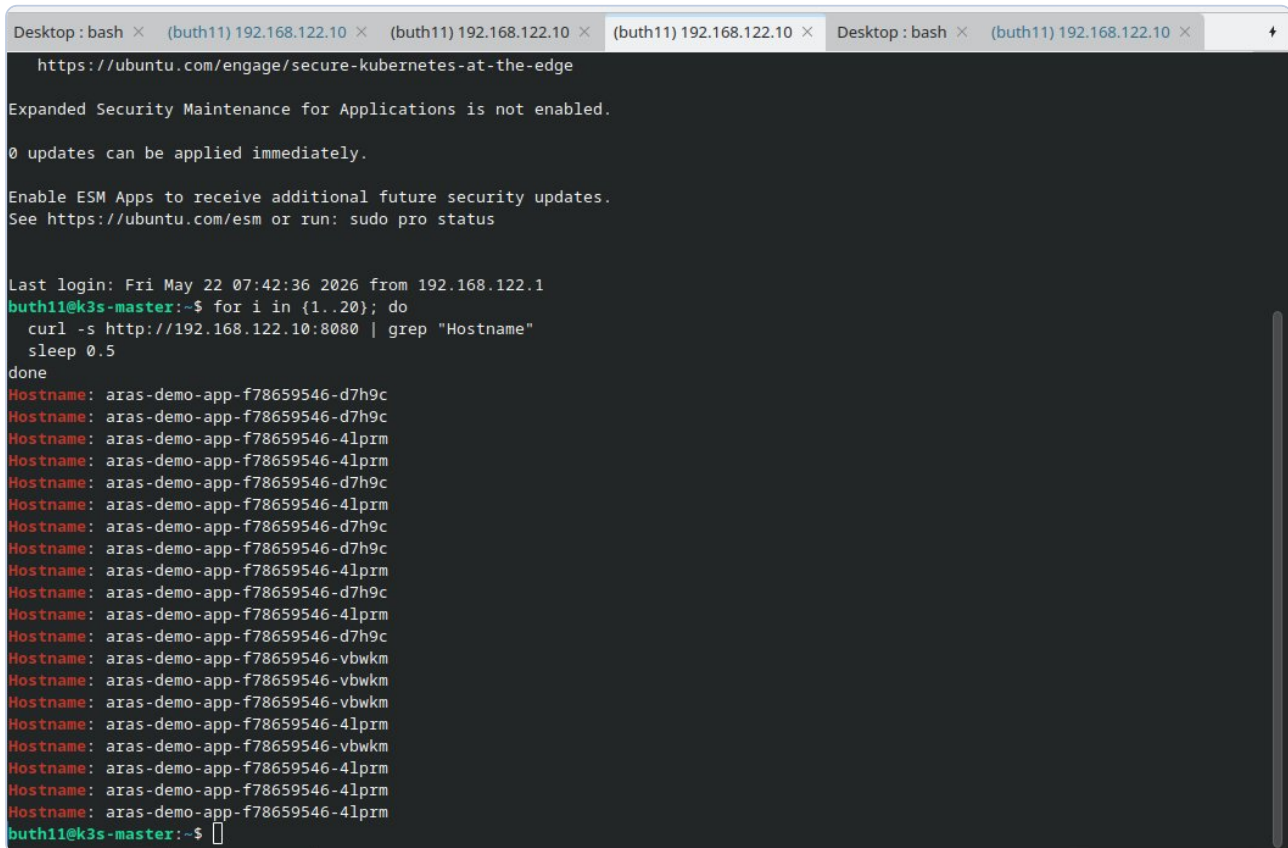
Computer	AvailabilityPct	ReadySamples	TotalSamples
k3s-master	100	184	184
Computer	k3s-master		
AvailabilityPct	100		
ReadySamples	184		
TotalSamples	184		
k3s-worker1	100	184	184
Computer	k3s-worker1		
AvailabilityPct	100		
ReadySamples	184		
TotalSamples	184		
k3s-worker2	100	184	184
Computer	k3s-worker2		
AvailabilityPct	100		
ReadySamples	184		
TotalSamples	184		

0s 582ms | Display time (UTC+00:00) | Query details | 1 - 3 of 3

KQL-5: SLO 100% — 184/184 samples Ready, error budget intact.

SLO 100% in 24h window. Enterprise target: 99.9% (43 min/month). Error budget intact even after full VM restart.

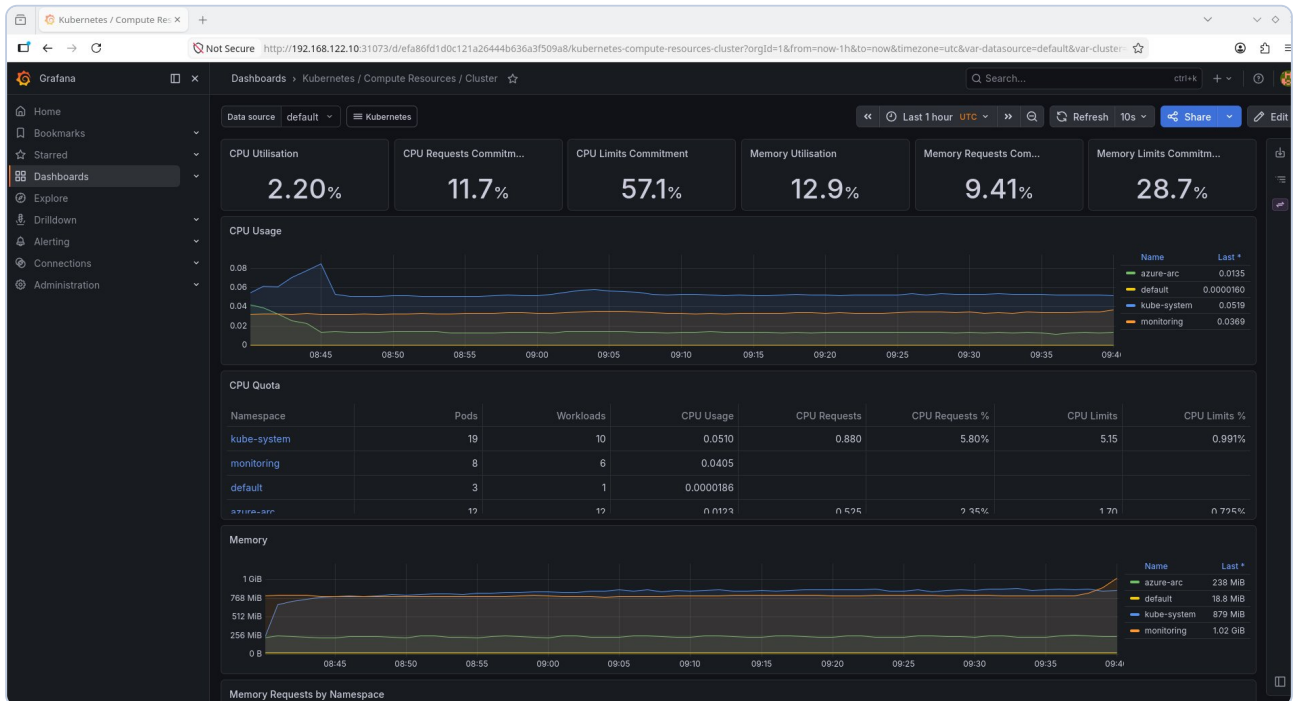
Chapter 7: Load Balancing



```
Desktop : bash × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × Desktop : bash × (buth11) 192.168.122.10 ×  
https://ubuntu.com/engage/secure-kubernetes-at-the-edge  
Expanded Security Maintenance for Applications is not enabled.  
0 updates can be applied immediately.  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
Last login: Fri May 22 07:42:36 2026 from 192.168.122.1  
buth11@k3s-master:~$ for i in {1..20}; do  
  curl -s http://192.168.122.10:8080 | grep "Hostname"  
  sleep 0.5  
done  
Hostname: aras-demo-app-f78659546-d7h9c  
Hostname: aras-demo-app-f78659546-d7h9c  
Hostname: aras-demo-app-f78659546-4lprm  
Hostname: aras-demo-app-f78659546-4lprm  
Hostname: aras-demo-app-f78659546-d7h9c  
Hostname: aras-demo-app-f78659546-4lprm  
Hostname: aras-demo-app-f78659546-d7h9c  
Hostname: aras-demo-app-f78659546-d7h9c  
Hostname: aras-demo-app-f78659546-4lprm  
Hostname: aras-demo-app-f78659546-d7h9c  
Hostname: aras-demo-app-f78659546-d7h9c  
Hostname: aras-demo-app-f78659546-4lprm  
Hostname: aras-demo-app-f78659546-d7h9c  
Hostname: aras-demo-app-f78659546-vbwkm  
Hostname: aras-demo-app-f78659546-vbwkm  
Hostname: aras-demo-app-f78659546-vbwkm  
Hostname: aras-demo-app-f78659546-4lprm  
Hostname: aras-demo-app-f78659546-vbwkm  
Hostname: aras-demo-app-f78659546-4lprm  
Hostname: aras-demo-app-f78659546-4lprm  
Hostname: aras-demo-app-f78659546-4lprm  
buth11@k3s-master:~$
```

Screenshot 5: Round-robin — 20 requests rotating across 3 pods, 900 parallel with zero errors.

Chapter 8: Prometheus & Grafana



Screenshot 6: Grafana — CPU 2.20%, RAM 12.9%, CPU Quota per namespace.

Chapter 9: Terraform IaC

```
Desktop: bash × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × terraform: bash × (buth11) 192.168.122.10 ×
Plan: 6 to add, 0 to change, 0 to destroy.

Changes to Outputs:
+ action_group_id           = (known after apply)
+ budget_name               = "SRE-Lab-Budget"
+ log_analytics_workspace_id = (known after apply)
+ log_analytics_workspace_key = (sensitive value)
+ resource_group_id         = (known after apply)
+ resource_group_name        = "SRE-Lab-RG"
+ subscription_id           = "b023e8b8-4446-4d4f-9fcf-d20a6c1fa093"

Saved the plan to: tfplan

To perform exactly these actions, run the following command to apply:
  terraform apply "tfplan"
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo/terraform$ terraform apply tfplan
random_string.suffix: Creating...
random_string.suffix: Creation complete after 0s [id=5plwbb]
azurerm_resource_group.sre_lab: Creating...
azurerm_consumption_budget_subscription.sre_lab: Creating...
azurerm_consumption_budget_subscription.sre_lab: Still creating... [00m10s elapsed]
azurerm_consumption_budget_subscription.sre_lab: Creation complete after 19s [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/providers/Microsoft.Consumption/budgets/SRE-Lab-Budget]

Error: A resource with the ID "/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG" already exists -
to be managed via Terraform this resource needs to be imported into the State. Please see the resource documentation for "azure
rm_resource_group" for more information.

with azurerm_resource_group.sre_lab,
on main.tf line 26, in resource "azurerm_resource_group" "sre_lab":
26: resource "azurerm_resource_group" "sre_lab" {

buth11@fedora:~/VMs/Azure/Github_repo/sre_repo/terraform$
```

Terraform Plan: 6 resources to create, 0 to change, 0 to destroy.

```
Desktop: bash × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × terraform: bash × (buth11) 192.168.122.10 ×

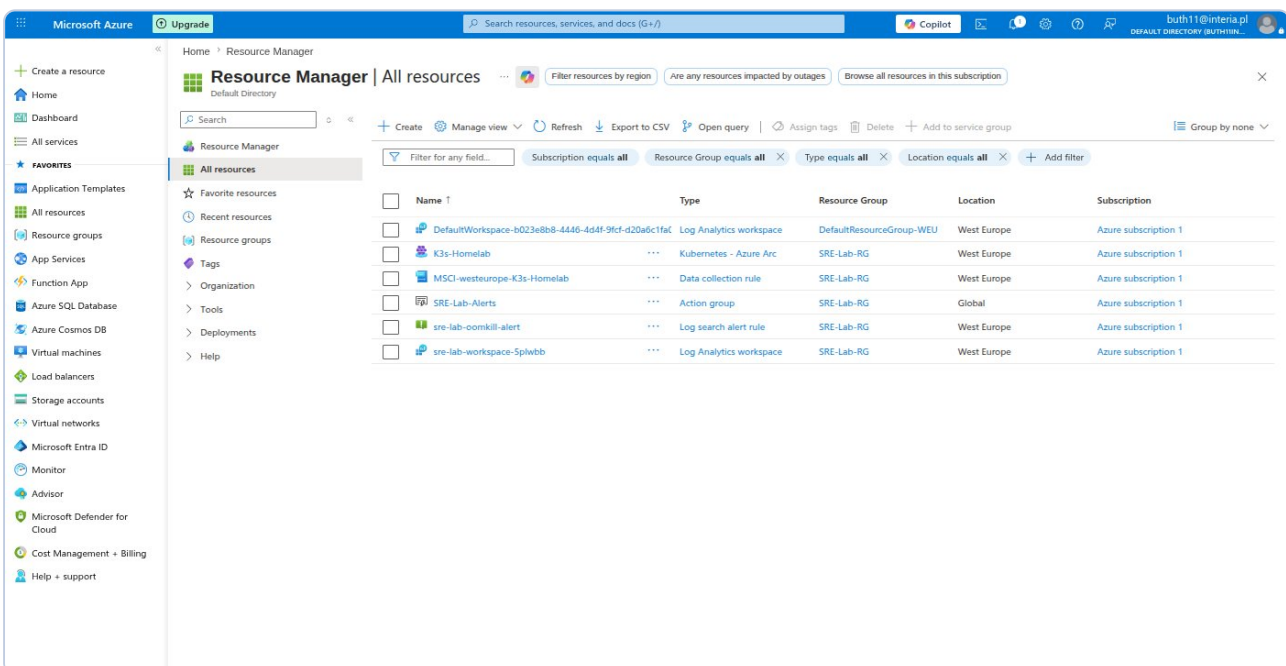
Saved the plan to: tfplan

To perform exactly these actions, run the following command to apply:
terraform apply "tfplan"
azurerm_resource_group.sre_lab: Modifying... [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG]
azurerm_resource_group.sre_lab: Modifications complete after 3s [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG]
azurerm_monitor_action_group.sre_alerts: Creating...
azurerm_log_analytics_workspace.sre_lab: Creating...
azurerm_monitor_action_group.sre_alerts: Creation complete after 4s [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.Insights/actionGroups/SRE-Lab-Alerts]
azurerm_log_analytics_workspace.sre_lab: Still creating... [00m10s elapsed]
azurerm_log_analytics_workspace.sre_lab: Still creating... [00m20s elapsed]
azurerm_log_analytics_workspace.sre_lab: Still creating... [00m30s elapsed]
azurerm_log_analytics_workspace.sre_lab: Still creating... [00m40s elapsed]
azurerm_log_analytics_workspace.sre_lab: Creation complete after 47s [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.OperationalInsights/workspaces/sre-lab-workspace-5plwbb]
azurerm_monitor_scheduled_query_rules_alert_v2.oom_kill: Creating...
azurerm_monitor_scheduled_query_rules_alert_v2.oom_kill: Creation complete after 5s [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.Insights/scheduledQueryRules/sre-lab-oomkill-alert]

Apply complete! Resources: 3 added, 1 changed, 0 destroyed.

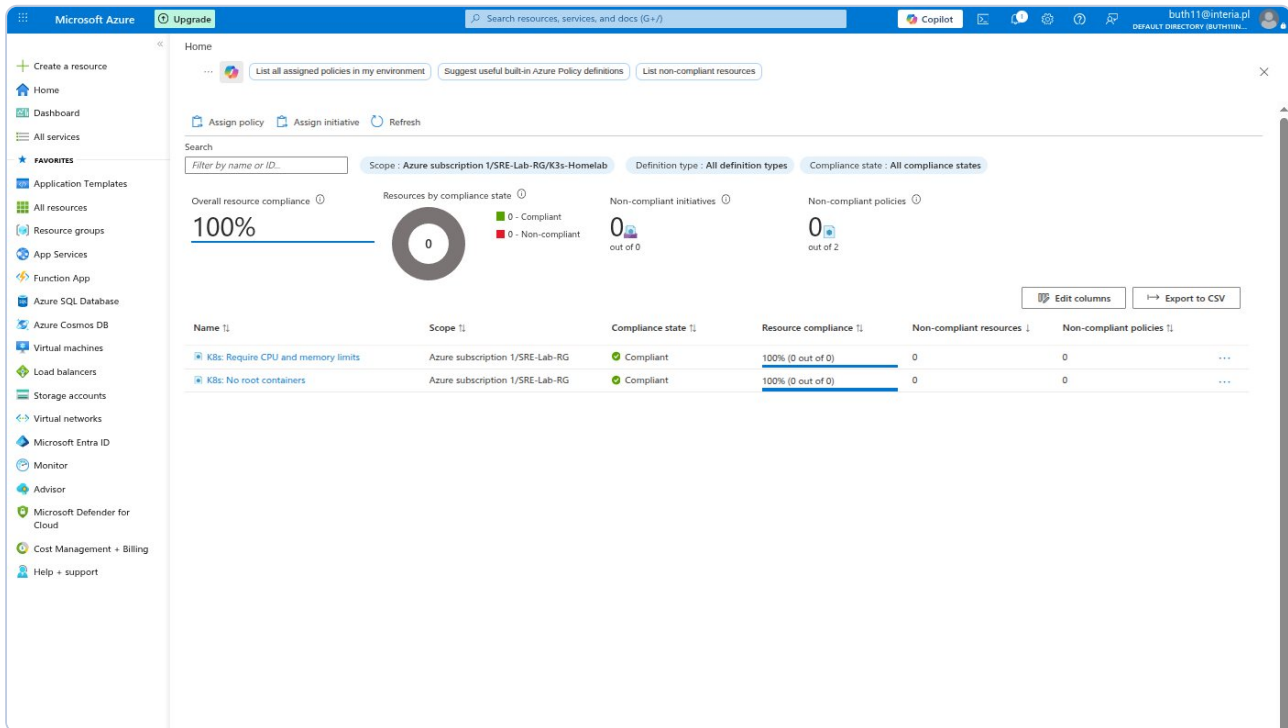
Outputs:
action_group_id = "/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.Insights/actionGroups/SRE-Lab-Alerts"
budget_name = "SRE-Lab-Budget"
log_analytics_workspace_id = "/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.OperationalInsights/workspaces/sre-lab-workspace-5plwbb"
log_analytics_workspace_key = <sensitive>
resource_group_id = "/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG"
resource_group_name = "SRE-Lab-RG"
subscription_id = "b023e8b8-4446-4d4f-9fcf-d20a6c1fa093"
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo/terraform$
```

Terraform Apply complete: 3 added, 1 changed, 0 destroyed.



Azure Portal: SRE-Lab-Alerts, sre-lab-oomkill-alert, sre-lab-workspace in SRE-Lab-RG.

Chapter 10: Azure Policy



Azure Policy: K8s No root containers — Compliant, K8s CPU/memory limits — Compliant, 100%.

Chapter 11: GitHub

The screenshot shows the GitHub repository page for 'sre-homelab-azure-arc' by user 'buth11'. The repository is public and has 1 branch (main) and 0 tags. The commit history shows an initial commit 'K3s + Azure Arc SRE Homelab' made 8 minutes ago. The file list includes folders 'docs', 'k8s', 'monitoring', 'scripts', and 'terraform', as well as files '.gitignore' and 'README.md'. The README file is selected, showing the title 'SRE Homelab: Azure Arc Hybrid Cloud Lab' and a description: 'A Proof-of-Value project demonstrating hybrid cloud orchestration, infrastructure-as-code, observability, and FinOps practices using K3s Kubernetes and Microsoft Azure Arc.' The right sidebar shows the repository's statistics: 0 stars, 0 forks, and 0 releases. The language distribution is shown as a bar chart: Shell 58.4% and HCL 41.6%.

GitHub repository page for **sre-homelab-azure-arc** (Public) by **buth11**.

Repository details: 1 Branch (main), 0 Tags, 1 Commit (Initial commit: K3s + Azure Arc SRE Homelab, 8 minutes ago).

Files and folders:

- docs
- k8s
- monitoring
- scripts
- terraform
- .gitignore
- README.md

README content:

SRE Homelab: Azure Arc Hybrid Cloud Lab

A Proof-of-Value project demonstrating hybrid cloud orchestration, infrastructure-as-code, observability, and FinOps practices using K3s Kubernetes and Microsoft Azure Arc.

Overview

Repository statistics:

- Readme
- Activity
- 0 stars
- 0 watching
- 0 forks
- Releases: No releases published. [Create a new release](#)
- Packages: No packages published. [Publish your first package](#)
- Contributors: 1 contributor (buth11)
- Languages: Shell 58.4%, HCL 41.6%

GitHub repo sre-homelab-azure-arc — public, Shell 58.4%, HCL 41.6%.

Chapter 12: Postmortem POST-001

Field	Value
Severity	P2
Duration	Under 7 minutes
Root Cause	Typo: ks3-worker1 instead of k3s-worker1
Status	Resolved — regex validation added to scripts

Week 1 Results

Metric	Value
Cluster nodes	3 / 3 Ready
Running pods	29
CPU utilisation	avg 2.20% / max 4.84%
RAM utilisation	avg 12.9% / max 14.48%
SLO Availability (24h)	100% — 184/184 samples Ready
Azure cost	0.00 USD
Terraform resources	5
Security policies	2 Compliant

Week 2: Service Principal, Remote State and CI/CD Pipeline

The second phase extends the infrastructure with elements critical for production environments: automated authentication without human interaction, centralized Terraform state storage with locking mechanism, and a fully automated CI/CD pipeline with an approval gate before deployment.

Service Principal — Automated Authentication

An Azure Service Principal is a technical account in Azure Active Directory — equivalent to a "service account" in on-premises systems. Instead of browser-based login with MFA, the pipeline uses an application ID and client secret stored as environment variables. The account has the Contributor role scoped to a specific subscription — least privilege principle.

```
az ad sp create-for-rbac --name "sre-homelab-sp" --role Contributor --scopes /subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093
```

The `ARM_CLIENT_ID`, `ARM_CLIENT_SECRET`, `ARM_TENANT_ID`, `ARM_SUBSCRIPTION_ID` variables are automatically read by Terraform and Azure CLI — eliminating the need for manual login on every execution.

Terraform Remote State in Azure Blob Storage

The local `terraform.tfstate` file was migrated to Azure Blob Storage. Remote state solves three production problems: file loss on disk failure, concurrent editing by multiple engineers (lock mechanism), and lack of change history (blob versioning).

Component	Value
Storage Account	sretfstate5496 (terraform-state-rg)
Container	tfstate
State file	sre-homelab.tfstate (14.53 KiB)
Encryption	Server-side encryption: true
Versioning	isVersioningEnabled: true
Lock mechanism	Azure Blob Lease (Acquiring/Releasing state lock)

```

buth11@fedora:~/VMs/Azure/Github_repo/sre_repo/terraform$ # Czekaj 30 sekund na propagację uprawnień w Azure AD
sleep 30

az storage blob list \
  --container-name tfstate \
  --account-name sretfstate5496 \
  --auth-mode login \
  --query "[].{name:name, size:properties.contentLength, modified:properties.lastModified}" \
  --output table

You do not have the required permissions needed to perform this operation.
Depending on your operation, you may need to be assigned one of the following roles:
  "Storage Blob Data Owner"
  "Storage Blob Data Contributor"
  "Storage Blob Data Reader"
  "Storage Queue Data Contributor"
  "Storage Queue Data Reader"
  "Storage Table Data Contributor"
  "Storage Table Data Reader"

If you want to use the old authentication method and allow querying for the right account key, please use the "--auth-mode" parameter and "key" value.

buth11@fedora:~/VMs/Azure/Github_repo/sre_repo/terraform$ # Pobiera klucz dostępu do Storage Account
# auth-mode key używa klucza zamiast OAuth - omija problem z uprawnieniami SP
STORAGE_KEY=$(az storage account keys list \
  --account-name sretfstate5496 \
  --resource-group terraform-state-rg \
  --query "[0].value" -o tsv)

# Listuje pliki w kontenerze używając klucza
az storage blob list \
  --container-name tfstate \
  --account-name sretfstate5496 \
  --account-key $STORAGE_KEY \
  --query "[].{name:name, size:properties.contentLength, modified:properties.lastModified}" \
  --output table

```

Name	Size	Modified
sre-homelab.tfstate	14881	2026-05-23T18:23:07+00:00

```

buth11@fedora:~/VMs/Azure/Github_repo/sre_repo/terraform$

```

Screenshot W2-1: Terminal — sre-homelab.tfstate (14881 bytes) visible in tfstate container after migration using storage account key.

The screenshot shows the Microsoft Azure portal interface. The left sidebar contains navigation options like 'Create a resource', 'Home', 'Dashboard', 'All services', and 'FAVORITES'. The main content area displays the properties of a Blob named 'sre-homelab.tfstate'. The 'Overview' tab is selected, showing a list of properties and their values. Below the properties, there are sections for 'Metadata' and 'Blob index tags' with input fields for keys and values.

Properties	
URL	https://sretfststate5496.blob...
LAST MODIFIED	5/23/2026, 8:23:07 PM
CREATION TIME	5/23/2026, 8:23:07 PM
VERSION ID	2026-05-23T18:23:07.9861361Z
TYPE	Block blob
SIZE	14.53 KiB
ACCESS TIER	Hot (Inferred)
ACCESS TIER LAST MODIFIED	N/A
ARCHIVE STATUS	-
REHYDRATE PRIORITY	-
SERVER ENCRYPTED	true
ETAG	0x8DEB8F856BD8961
VERSION-LEVEL IMMUTABILITY POLICY	Disabled
CACHE-CONTROL	
CONTENT-TYPE	application/json
CONTENT-MD5	2mHfbTY53lmCzqo/RWS+MQ...
CONTENT-ENCODING	
CONTENT-LANGUAGE	
CONTENT-DISPOSITION	
LEASE STATUS	Unlocked
LEASE STATE	Available
LEASE DURATION	-
COPY STATUS	-
COPY COMPLETION TIME	-

Below the properties, there is a blue 'Undelete' button. The 'Metadata' section has a table with 'Key' and 'Value' columns. The 'Blob index tags' section also has a table with 'Key' and 'Value' columns.

Screenshot W2-2: Azure Portal — blob properties: SIZE 14.53 KiB, SERVER ENCRYPTED true, VERSION ID active, LEASE STATUS Unlocked/Available.

```

buth11@fedora:~/VMs/Azure/Github_repo/sre_repo/terraform$ # terraform plan pobierze state z Azure Blob zamiast lokal
nego pliku
# Powinno pokazac "No changes" bo infrastruktura jest aktualna
terraform plan
Acquiring state lock. This may take a few moments...
random_string.suffix: Refreshing state... [id=5plwbb]
data.azurearm_subscription.current: Reading...
azurearm_resource_group.sre_lab: Refreshing state... [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resource
Groups/SRE-Lab-RG]
azurearm_monitor_action_group.sre_alerts: Refreshing state... [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093
/resourceGroups/SRE-Lab-RG/providers/Microsoft.Insights/actionGroups/SRE-Lab-Alerts]
azurearm_log_analytics_workspace.sre_lab: Refreshing state... [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093
/resourceGroups/SRE-Lab-RG/providers/Microsoft.OperationalInsights/workspaces/sre-lab-workspace-5plwbb]
data.azurearm_subscription.current: Read complete after 1s [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093]
azurearm_consumption_budget_subscription.sre_lab: Refreshing state... [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a
6c1fa093/providers/Microsoft.Consumption/budgets/SRE-Lab-Budget]
azurearm_monitor_scheduled_query_rules_alert_v2.oom_kill: Refreshing state... [id=/subscriptions/b023e8b8-4446-4d4f-9
fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.Insights/scheduledQueryRules/sre-lab-oomkill-alert]

No changes. Your infrastructure matches the configuration.

Terraform has compared your real infrastructure against your configuration and found no differences, so no changes
are needed.
Releasing state lock. This may take a few moments...
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo/terraform$

```

Screenshot W2-3: terraform plan with remote state — "Acquiring state lock... No changes. Releasing state lock." Terraform fetches state from Azure Blob instead of local file.

Key evidence: "Acquiring state lock" and "Releasing state lock" lines in terraform plan output confirm the locking mechanism works. In a multi-engineer environment, a second engineer cannot run apply while the first is already executing it.

Azure DevOps CI/CD Pipeline

The pipeline defined in YAML (pipelines/terraform-ci.yml) automatically triggers on every push of changes to the terraform/ directory. It consists of two stages with clear separation of responsibilities.

Stage	Steps	Duration	Trigger
Terraform Validate & Plan	install, init, validate, plan, publish artifact	37s	Automatic on push
Terraform Apply	init, apply (from artifact)	49s	Manual approval gate

Screenshot W2-4: Azure DevOps Library — Variable Group "terraform-credentials" with 4 ARM_* variables. ARM_CLIENT_SECRET hidden (lock icon) — secrets never visible in pipeline logs.

Azure DevOps

buth11 / sre-homelab-arc / Pipelines / Environments / production

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sre-homelab-arc

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Approvals and checks

Check execution order

Checks will run by order, which is pre-determined by check types. Checks within the same order will run in parallel.
[Learn more](#)

+

Add new

Order	Display name	Type	Timeout	Details
1	All approvers must approve	Approvals	30d	BS

Screenshot W2-5: Environment "production" — Approval Gate: "All approvers must approve", Timeout 30 days. Pipeline stops before apply and sends notification to engineer.

Bartosz Suszko | SRE Case Study | 25 / 31

Azure DevOps Interface showing the 'Jobs in run' section for the pipeline 'buth11.sre-homelab-azure-arc' with ID #20260523.1. The jobs listed are:

Job Name	Status	Duration
terraform plan	Completed	34s
Initialize job	Completed	<1s
Checkout buth1...	Completed	2s
Install Terraform...	Completed	2s
terraform init	Completed	7s
terraform validate	Completed	1s
terraform plan	Completed	15s
Publish terrafor...	Completed	3s
Post-job: Chec...	Completed	<1s
Finalize Job	Completed	<1s

The 'terraform plan' job is expanded, showing the following log output:

```
1 Agent: Azure Pipelines 1
2 Started: Just now
3 Duration: 34s
4
5 Job preparation parameters
44 1 artifact produced
45 Job live console data:
46 Starting: terraform plan
47 Async Command Start: DetectDockerContainer
48 Async Command End: DetectDockerContainer
49 Async Command Start: DetectDockerContainer
50 Async Command End: DetectDockerContainer
51 Finishing: terraform plan
```

Screenshot W2-6: Stage 1 complete — all steps green: Install Terraform 1.15.4 (2s), terraform init (7s), terraform validate (1s), terraform plan (15s), Publish artifact (3s). 1 artifact produced.

Azure DevOps | **sre-homelab-arc** | 20260523.1

#20260523.1 • feat: add Azure DevOps CI/CD pipeline for Terraform

Summary | Code Coverage

Manually run by **Bartosz Suszko** | View change | Parameters

Today at 9:13 PM | 1m 45s

Repository and version | **buth11/sre-homelab-azure-arc** | main | e9da4bfb

Related | 0 work items | 1 published

Tests and coverage | Get started

This pipeline needs permission to access a resource before this run can continue to Terraform Apply | View

Stages | Jobs

Terraform Validate ... | 1 job completed | 37s | 1 artifact

Terraform Apply | Waiting | Permission needed

Project settings

Screenshot W2-7: Pipeline waiting for approval — Stage 1 green (37s, 1 artifact), Stage 2 "Waiting" with "1 approval needs your review". Approve/Reject buttons visible.

The screenshot displays the Azure DevOps web interface for a project named 'sre-homelab-arc'. The left sidebar shows navigation options: Overview, Boards, Repos, Pipelines (selected), Environments, Releases, Library, Task groups, Deployment groups, Test Plans, and Artifacts. The main area shows a pipeline run #20260523.1 with the title 'feat: add Azure DevOps CI/CD pipeline for Terraform'. A status bar indicates the run is being retained. Below this, a 'Summary' tab is active, showing the run was manually triggered by Bartosz Suszko at 9:13 PM, taking 4m 24s. It lists the repository as 'buth11/sre-homelab-azure-arc' on the 'main' branch with commit 'e9da4bfb'. The run includes 0 work items and 1 published item. The 'Stages' section shows two jobs: 'Terraform Validate ...' (37s, 1 artifact) and 'Terraform Apply' (49s, 1 checks passed), both marked as successful with green checkmarks.

Screenshot W2-8: Pipeline completed successfully — Terraform Validate & Plan (37s, 1 artifact) and Terraform Apply (49s, 1 checks passed) both green. Total time: 4m 24s.

On interview: "Every push to terraform/ triggers an automatic plan. Apply requires manual engineer approval after plan review — exactly as in enterprise environments where infrastructure changes must be approved before production deployment."

Troubleshooting Log

ISSUE-001 KQL: Failed to resolve scalar expression CPUCapacityNanoCores

```
project operator: Failed to resolve scalar expression named  
CPUCapacityNanoCores
```

Root Cause: AMA v3.x uses different column names than documentation for older omsagent. Column does not exist in this schema version.

Resolution: Ran KubeNodeInventory | getschema and simplified query to available columns.

ISSUE-002 Azure Policy: MissingPolicyParameter cpuLimit, memoryLimit

```
(MissingPolicyParameter) policy assignment k8s-resource-limits is missing  
cpuLimit, memoryLimit
```

Root Cause: Policy definition version 9.3.0 introduced mandatory parameters not required in earlier versions.

Resolution: Added --params with cpuLimit: 500m and memoryLimit: 512Mi to CLI command.

ISSUE-003 Terraform: Resource Group already exists (brownfield import)

Error: A resource with the ID .../SRE-Lab-RG already exists – needs to be imported into State

Root Cause: Resource Group was previously created manually by az connectedk8s connect. Terraform had no state entry for it.

Resolution: terraform import azure_rm_resource_group.sre_lab /subscriptions/.../SRE-Lab-RG

ISSUE-004 Terraform: Saved plan is stale after import

Error: Saved plan is stale – state was changed by another operation after plan was created

Resolution: terraform plan -out=tfplan and terraform apply tfplan after import.

ISSUE-005 Container Insights: Extension missing after VM restart

Root Cause: Cluster identity certificate was regenerated on restart — AMA agents required re-registration.

Resolution: az k8s-extension create --name azuremonitor-containers — agents returned to Running in 37 seconds.

ISSUE-006 POST-001: K3s etcd Split-Brain (hostname typo: ks3 vs k3s)

FATA[0034] Node password rejected, duplicate hostname

Root Cause: Hostname typo ks3-worker1 instead of k3s-worker1 during VM provisioning. etcd treats node identity as immutable.

Resolution: kubectl delete node + rm /etc/rancher/node/ + restart k3s-agent.
Resolution time: 7 minutes. Scripts updated with regex validation.

ISSUE-007 Azure DevOps: TerraformInstaller task not found

A task is missing: TerraformInstaller. This task isn't installed and may be available from Marketplace.

Root Cause: Extension ms-devlabs.custom-terraform-tasks is not installed by default in Azure DevOps organizations.

Resolution: Installed from Marketplace: marketplace.visualstudio.com/items?itemName=ms-devlabs.custom-terraform-tasks — Get it free, select both org, Install.

ISSUE-008 Azure DevOps: Pipeline needs permission to access Variable Group

This pipeline needs permission to access a resource before this run can continue to Terraform Validate & Plan

Root Cause: First pipeline run requires one-time authorization to access Variable Group and production Environment.

Resolution: Clicked View > Permit in the authorization window. One-time only — all future runs proceed automatically.